

Ref No.: CECNF/N/074

Roll No. _____

B.A./B.Sc. (Hons.) Semester III Examination 2025-26 (NEP)**STATISTICS****STAMV 31: Connecting Data with Probability Models**

Time: 2:30 Hours

Max. Marks: 70

Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.

Note: Attempt any five questions. Each question carries equal marks. Question no. 1 is compulsory.

1.

- (a) Consider the following situations and identify which one requires repetition of the experiment to determine the required probability. Give reasons for your answer.
- (i) An unbiased coin is tossed once. What is probability that a head will come on top?
 - (ii) A drug is administered to a patient to induce sleep. What is the probability that the patient falls asleep within 10 minutes?
 - (iii) A coin is tossed once. What is probability that a head will come on top?
- (b) Name the appropriate probability model (distribution) applicable in each of the following situations. Give brief justification for your choice.
- (i) A student appears repeatedly in the UPSC examination until he qualifies and records the number of failures before the first success.
 - (ii) A couple plans to have four children and counts the number of male children in the family.
 - (iii) From a group of 10 males and 5 females, 4 individuals are selected at random without replacement, and the number of males in the selected sample is recorded.
- (c) Differentiate between:
- (i) Population and Sample
 - (ii) Parameter and statistic
- (d) Define following terms with example:
- (i) Null hypothesis
 - (ii) Alternate hypothesis
 - (iii) Type I error
 - (iv) Type II error

[14]

2.

(a) Write axioms of probability. Using the axioms show that for any two events A and B:

(i) $P(A) = 1 - P(A^C)$

(ii) $P(\emptyset) = 0$

(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(iv) $P[(A \cap B^C) \cup (A^C \cap B)] = P(A) + P(B) - 2P(A \cap B)$

(b) Define conditional probability and independence of events. There are 3 defective and 4 good items in a bag. Two items are being chosen at random. Let $A = \{\text{First item selected is defective}\}$ and $B = \{\text{2nd selected item is defective}\}$. $P(A) = P(B)$ irrespective of selection methods with and without replacement.

[14]

3.

(a) Define Bernoulli distribution. Explain a situation where this distribution can be fitted. Let $X \sim \text{bern}(p)$ then show that first four raw moments of the distribution are same.

(b) Let $X_1, X_2, \dots, X_n$ are identically and independently distributed random variables from Bernoulli distribution with parameter p . If $Y = \sum_{i=1}^n X_i$, then find probability distribution of Y and obtain $E(Y^2)$.

[14]

4.

(a) In a study, the number of male children was recorded in 100 families, each having four children. The observed frequencies are given below:

Number of male children	0	1	2	3	4	Total
Number of families	6	25	38	25	6	100

Assuming a binomial model, examine whether the distribution of male children per family is fit to binomial and whether the probabilities of male and female births are equal.

(b) Define geometric distribution and obtain its mean.

[14]

5.

- (a) Show that following function defined for a random variable is a probability density function:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{if } -\infty < x < \infty$$

Using distribution function approach obtain Pdf of $Y = X^2$ and identify (by name) the probability distribution of Y .

- (b) A population consists of five values 1, 2, 3, 4 and 5. First, sketch the frequency distribution of the population data and draw its corresponding graph. Next, draw all possible samples of size 2 from the population using the with-replacement method. For each sample, compute the sample mean and then prepare the frequency distribution of these sample means. Draw a graph of the frequency distribution of the sample means. Finally, compare the two graphs and comment on your observations. Which of the two-frequency distribution represents sampling distribution?

[14]

6.

- (a) Explain the maximum likelihood estimation method.
- (b) Let $x_1, x_2, \dots, x_n$ be a random sample from a population $X \sim N(\mu, \sigma_0^2)$ where σ_0^2 is known. Show that the sample mean $\bar{x}$ is an estimator of μ by the method of maximum likelihood as well as moment.

[14]

7.

- (a) Show that the sampling distribution of sample means $\bar{x}$ in case of a normal population $X \sim N(\mu, \sigma_0^2)$, where σ_0^2 is known, is $N\left(\mu, \frac{\sigma_0^2}{n}\right)$.
- (b) If $x \geq 1$, is the critical region for testing $H_0 : \theta = 2$ against the alternative $H_1 : \theta = 1$ on the basis of the single observation from the population:

$$f(x, \theta) = \theta e^{-x\theta}, \quad 0 \leq x \leq \infty, \theta \geq 0$$

obtain the values of type I and type II errors.

[14]

[Click here to visit our website](#)

[Click here to visit our contributor](#)